

Seyedreza Rezaei

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EDUCATION

York University

Ph.D. Candidate in Robotics and Aerospace Engineering

- Department of Earth and Space Science and Engineering | GPA: 4.0/4.0

Toronto, Canada

Sep 2023 – Present

University of Rome Tor Vergata

M.Sc. Mechatronics Engineering

- Department of Electronic Engineering | Final GPA: 104/110 (Top 3 Student)

Rome, Italy

Sep 2019 – Oct 2022

University of Stuttgart

Erasmus+ Exchange Student

- Specialized in Deep Learning, Nonlinear and Model Predictive Control

Stuttgart, Germany

Mar 2021 – Aug 2021

Shiraz University

B.Sc. Mechanical Engineering

- School of Mechanical Engineering | Final GPA: 14.62/20 (Field's Average: 13.56/20)

Shiraz, Iran

Sep 2015 – Feb 2019

EXPERIENCE

Lassonde School of Engineering, York University

Teaching Assistant & Research Assistant

- Teaching Assistant for PHYS 1411, PHYS 1412, PHYS 1800, PHYS 1801, and NATS 1740.
- Doctoral researcher at the Spacecraft Dynamics, Control and Navigation Laboratory (SDCNLab).
- Conducted simulation and experimental validation on UAV platforms.

Toronto, Canada

Sep 2023 – Present

Quanser

Ph.D. Researcher Intern

- Industry-funded research internship supported by the Mitacs Accelerate program.
- Contributed to development and testing of the QDrone 2 platform.
- Implemented and validated control algorithms on hardware.

Toronto, Canada

Sep 2023 – May 2024

Max Planck Institute for Intelligent Systems

Researcher

- Full-time researcher in the Locomotion in Biorobotic and Somatic Systems Group.
- Master's thesis on repetitive learning control for soft robotic fish.
- Designed and conducted experimental validation studies.

Stuttgart, Germany

Feb 2022 – Feb 2023

Laboratory of C-PED (Center for Power Electronics and Drives)

Research Assistant

- Developed FPGA-based nonlinear control for active rectifiers.
- Performed hardware testing and validation.

Rome, Italy

Sep 2020 – Feb 2021

Shiraz University, School of Mechanical Engineering

Research Assistant

- Research on structural vibration, nano-mechanics, and nano-bio materials.
- Numerical modeling and experimental analysis.

Shiraz, Iran

Jul 2018 – Feb 2019

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, C++, Bash, ROS.
- **Control & Robotics:** Nonlinear MPC, Control Barrier Functions, Passivity-Based Control, State Estimation.
- **Optimization & Simulation:** CasADi, ACADOS, Simulink, Gazebo.
- **Design & Documentation:** SolidWorks, $\LaTeX$, Microsoft Office, NI LabVIEW.

PUBLICATIONS

- **Rezaei, S.**, Kang, J., Haridevan, A., Shan, J., “SEP-NMPC: Safety Enhanced Passivity-Based Nonlinear Model Predictive Control for a UAV Slung Payload System,” accepted at *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- **Rezaei, S.**, Kang, J., Shan, J., “Hierarchical NMPC with Constraint Decomposition for Safe and Stable UAV Slung-load Transportation,” in preparation, planned submission to *IEEE Robotics and Automation Letters (RA-L)*, Summer 2026.
- Yousaf, A., Doostmohammadi, A., **Rezaei, S.**, Rezai, P., Shan, J., “Real-Time Detection of Toluene in Aqueous Headspace Using a Low-Cost UAV-Compatible Gas Sensor Platform,” Manuscript in preparation.
- Schulz, A. K., Chellapurath, M., Khandelwal, P. C., **Rezaei, S.**, Merker, S., Jusufi, A., “Scaly-tail organ enhances static stability during Pel’s scaly-tailed flying squirrels’ arboreal locomotion,” *Journal of the Royal Society Interface*, vol. 22, no. 227, 2025, Art. 20240937.
- Schwab, F., El Arayshi, M., **Rezaei, S.**, Sprumont, H., Allione, F., Mucignat, C., Lunati, I., Verrelli, C. M., Jusufi, A., “Learning control for body caudal undulation with soft sensory feedback,” *Frontiers in Sensors*, vol. 5, 2024, Art. 1367992.

HONORS AND AWARDS

- **Mitacs Accelerate Internship Award** — Supported research internship at Quanser through the Mitacs Accelerate program, a competitive research-based industry internship funding opportunity for graduate students.
- **Ranked 1st** in M.Sc. Mechatronics Engineering for two years and ranked among the top 3 students overall.
- **Erasmus+ Scholarship** for study at the University of Stuttgart (2020–21) and Erasmus+ Traineeship for research at the Max Planck Institute for Intelligent Systems (2021–22).
- **DSU Borsista** (Diritto Allo Studio Universitario) for M.Sc. study from the Government of Italy.
- **Student Grants** for a Research Assistant position at Shiraz University from Prof. Esmaeel Ghvanloo.
- **Governmental Tuition-Waiving Fellowship** for Bachelor study from Iran’s Ministry of Science.

RESEARCH COMMUNICATION & GRANT CONTRIBUTIONS

- **CSME Bulletin (Fall 2025)**: “Planning and Control of UAV-Slung-Load Systems for Efficient and Smarter Aerial Transportation,” Canadian Society for Mechanical Engineering, Special Issue on Advances in Aerospace.
- **Swiss National Science Foundation**: “Soft Paleo-Robotics: Recreating Ancient Marine Reptiles’ Agile Swimming,” Grant Co-Writer (PI: Dr. Ardian Jusufi).
- **Swiss National Science Foundation Starting Grant**: “Soft Paleo-Robotics,” Grant Co-Writer (PI: Dr. Ardian Jusufi).
- **Joint Proposal Submission (D-A-CH)**: “Obstacle Traversal with Soft Active Appendages,” Grant Co-Writer (PI: Dr. Ardian Jusufi).

REFERENCES

- **Prof. Jinjun Shan**: Professor, Department of Earth and Space Science and Engineering, Lassonde School of Engineering, York University, Toronto, Canada.
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- **Prof. Ryan Orszulik**: Associate Professor, Department of Earth and Space Science and Engineering, Lassonde School of Engineering, York University, Toronto, Canada.
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- **Prof. Cristiano Maria Verrelli**: Associate Professor, Department of Electronic Engineering, University of Rome Tor Vergata, Rome, Italy.
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